Structural Regression & Mediation

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1. {lavaan} Basics

1.1. Overview of {lavaan}

The {lavaan} package provides functions for conducting various latent variable modeling techniques, including path analysis, confirmatory factor analysis, structural equation modeling, and growth curve modeling.

The 'official' tutorial is available online, and a downloadable PDF version of the tutorial is also available here. The tutorial utilizes illustrative examples to introduce the fundamental elements of {lavaan}: model syntax, fitting functions (e.g., `cfa`, `sem`, `growth`), and essential extractor functions (e.g., `summary`, `coef`, `fitted`, `inspect`). Additionally, it covers important topics such as mean structure, multiple groups, growth curve modeling, mediation analysis, and categorical data.

1.2. Setting up Environment

```
### Set the 'Cloud' mirror as a default CRAN (Comprehensive R Archive Network) repository
### CRAN is a global network of web servers that store identical, up-to-date versions of code and docum
options(repos="https://cloud.r-project.org")
```

2. Latent Regression

Latent regression is a statistical modeling technique that combines elements of latent variable modeling and regression analysis. It is used to explore the relationship between observed variables (often referred to as manifest variables) and unobserved variables (often referred to as latent variables or factors).

2.1. Specify a Measurement Model

A measurement model is a fundamental component of the overall SEM framework. The measurement model represents the relationships between latent variables and their observed indicators (manifest variables). It is used to describe how the latent variables are measured by the observed variables and to account for measurement error.

Once the measurement model is established and assessed, it serves as the foundation for the structural model in SEM, which examines the relationships between latent variables and predicts outcomes or dependent variables.

```r
### Read the SPSS data
#install.packages("haven")
library(haven)

panas7 <- read_spss("panas7.sav")
head(panas7)
```

```
## # A tibble: 6 x 19
##   agency1 agency2 agency3 intrin1 intrin2 intrin3 extrin1 extrin2 extrin3
##     <dbl>   <dbl>   <dbl>   <dbl>   <dbl>   <dbl>   <dbl>   <dbl>   <dbl>
## 1    3.5     4       4       4        4         4    1       1       1.5
## 2    2.5     3.17    3       3.21     2         3    1.83    2.67    1.83
## 3    1.83    2       1.5     3        3         2    1       1       1
## 4    2.77    3.06    2.36    3.13     4         3    1.08    1.17    1
## 5    3.17    3.33    2.83    3.5      4         4    1.83    2       1.83
## 6    2.33    2.83    2.33    3        2.5       3    3.06    2.41    2.67
## # i 10 more variables: posaff1 <dbl>, posaff2 <dbl>, posaff3 <dbl>,
## #   negaff1 <dbl>, negaff2 <dbl>, negaff3 <dbl>, female <dbl+lbl>,
## #   black <dbl+lbl>, hispanic <dbl+lbl>, other <dbl+lbl>
```

```r
### Measurement model
install.packages("lavaan")
```

```
## Installing package into 'C:/Users/Jason Lee/AppData/Local/R/win-library/4.3'
## (as 'lib' is unspecified)
```

```
##
##   There is a binary version available but the source version is later:
##        binary source needs_compilation
## lavaan 0.6-19 0.6-21             FALSE
```

```
## installing the source package 'lavaan'
```

```r
library(lavaan)
```

```
## This is lavaan 0.6-21
## lavaan is FREE software! Please report any bugs.
```

```r
m1 <- 'Agency =~ agency1 + agency2 + agency3
       Intrin =~ intrin1 + intrin2 + intrin3
       Extrin =~ extrin1 + extrin2 + extrin3
       PosAff =~ posaff1 + posaff2 + posaff3'
out1 <- cfa(model = m1, data = panas7, std.lv = TRUE, meanstructure = TRUE)
summary(out1, standardized = TRUE, fit.measures = TRUE, rsquare = TRUE)
```

```
## lavaan 0.6-21 ended normally after 54 iterations
##
##   Estimator                                         ML
##   Optimization method                           NLMINB
##   Number of model parameters                        42
##
##   Number of observations                           380
##
## Model Test User Model:
##
##   Test statistic                                57.525
##   Degrees of freedom                                48
##   P-value (Chi-square)                           0.163
##
## Model Test Baseline Model:
##
##   Test statistic                              2806.169
##   Degrees of freedom                                66
##   P-value                                        0.000
##
## User Model versus Baseline Model:
##
##   Comparative Fit Index (CFI)                    0.997
##   Tucker-Lewis Index (TLI)                       0.995
##
## Loglikelihood and Information Criteria:
##
##   Loglikelihood user model (H0)              -2976.044
##   Loglikelihood unrestricted model (H1)      -2947.282
##
##   Akaike (AIC)                                6036.088
##   Bayesian (BIC)                              6201.576
##   Sample-size adjusted Bayesian (SABIC)       6068.318
##
## Root Mean Square Error of Approximation:
##
##   RMSEA                                          0.023
##   90 Percent confidence interval - lower         0.000
##   90 Percent confidence interval - upper         0.042
##   P-value H_0: RMSEA <= 0.050                    0.992
##   P-value H_0: RMSEA >= 0.080                    0.000
##
## Standardized Root Mean Square Residual:
##
##   SRMR                                           0.026
##
```

```
## Parameter Estimates:
##
##   Standard errors                             Standard
##   Information                                 Expected
##   Information saturated (h1) model          Structured
##
## Latent Variables:
##                    Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
##   Agency =~
##     agency1           0.466    0.021   22.559    0.000    0.466    0.905
##     agency2           0.492    0.021   23.764    0.000    0.492    0.933
##     agency3           0.497    0.022   22.826    0.000    0.497    0.911
##   Intrin =~
##     intrin1           0.539    0.039   13.918    0.000    0.539    0.702
##     intrin2           0.579    0.043   13.435    0.000    0.579    0.681
##     intrin3           0.618    0.038   16.272    0.000    0.618    0.803
##   Extrin =~
##     extrin1           0.386    0.022   17.454    0.000    0.386    0.804
##     extrin2           0.456    0.027   17.116    0.000    0.456    0.791
##     extrin3           0.473    0.026   18.204    0.000    0.473    0.831
##   PosAff =~
##     posaff1           0.570    0.028   20.157    0.000    0.570    0.854
##     posaff2           0.604    0.029   21.175    0.000    0.604    0.883
##     posaff3           0.631    0.029   21.600    0.000    0.631    0.894
##
## Covariances:
##                    Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
##   Agency ~~
##     Intrin            0.507    0.047   10.795    0.000    0.507    0.507
##     Extrin            0.270    0.054    5.035    0.000    0.270    0.270
##     PosAff            0.302    0.051    5.947    0.000    0.302    0.302
##   Intrin ~~
##     Extrin           -0.027    0.063   -0.433    0.665   -0.027   -0.027
##     PosAff            0.413    0.052    7.910    0.000    0.413    0.413
##   Extrin ~~
##     PosAff           -0.029    0.058   -0.494    0.621   -0.029   -0.029
##
## Intercepts:
##                    Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
##    .agency1           2.442    0.026   92.349    0.000    2.442    4.737
##    .agency2           2.550    0.027   94.360    0.000    2.550    4.841
##    .agency3           2.544    0.028   90.926    0.000    2.544    4.664
##    .intrin1           3.002    0.039   76.152    0.000    3.002    3.907
##    .intrin2           2.987    0.044   68.467    0.000    2.987    3.512
##    .intrin3           3.080    0.039   78.022    0.000    3.080    4.002
##    .extrin1           1.615    0.025   65.603    0.000    1.615    3.365
##    .extrin2           1.686    0.030   57.065    0.000    1.686    2.927
##    .extrin3           1.633    0.029   55.873    0.000    1.633    2.866
##    .posaff1           3.136    0.034   91.663    0.000    3.136    4.702
##    .posaff2           2.991    0.035   85.211    0.000    2.991    4.371
##    .posaff3           3.069    0.036   84.776    0.000    3.069    4.349
##
## Variances:
##                    Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
```

```
##    .agency1           0.048    0.005    9.650    0.000    0.048    0.181
##    .agency2           0.036    0.005    7.622    0.000    0.036    0.129
##    .agency3           0.050    0.005    9.260    0.000    0.050    0.169
##    .intrin1           0.300    0.029   10.237    0.000    0.300    0.507
##    .intrin2           0.388    0.036   10.662    0.000    0.388    0.536
##    .intrin3           0.210    0.029    7.346    0.000    0.210    0.356
##    .extrin1           0.081    0.009    9.023    0.000    0.081    0.354
##    .extrin2           0.124    0.013    9.447    0.000    0.124    0.374
##    .extrin3           0.101    0.013    7.996    0.000    0.101    0.310
##    .posaff1           0.120    0.012    9.890    0.000    0.120    0.270
##    .posaff2           0.103    0.012    8.575    0.000    0.103    0.221
##    .posaff3           0.100    0.013    7.939    0.000    0.100    0.201
##     Agency            1.000                               1.000    1.000
##     Intrin            1.000                               1.000    1.000
##     Extrin            1.000                               1.000    1.000
##     PosAff            1.000                               1.000    1.000
##
## R-Square:
##                    Estimate
##     agency1           0.819
##     agency2           0.871
##     agency3           0.831
##     intrin1           0.493
##     intrin2           0.464
##     intrin3           0.644
##     extrin1           0.646
##     extrin2           0.626
##     extrin3           0.690
##     posaff1           0.730
##     posaff2           0.779
##     posaff3           0.799
```

```r
install.packages("lavaanPlot")
```

```
## Installing package into 'C:/Users/Jason Lee/AppData/Local/R/win-library/4.3'
## (as 'lib' is unspecified)
```

```
## package 'lavaanPlot' successfully unpacked and MD5 sums checked
##
## The downloaded binary packages are in
##  C:\Users\Jason Lee\AppData\Local\Temp\RtmpWW9w36\downloaded_packages
```

```r
library(lavaanPlot)
lavaanPlot(model = out1,
           node_options = list(shape = "box", fontname = "Helvetica"),
           coefs = TRUE, # Show coefficients
           covs = TRUE,  # Show covariances/correlations
           stand = TRUE, # Use standardized coefficients
           stars = "latent") # Show significance stars
```

2.2. Specify a Structural Model (Latent Regression)

A structural model is a component of the overall SEM framework that focuses on examining the relationships between latent variables and predicting outcomes or dependent variables. While the measurement model deals with how latent variables are measured by their observed indicators, the structural model explores the relationships among the latent variables themselves and between latent variables and observed variables.

```r
### Add latent regression paths
m2 <- 'Agency =~ agency1 + agency2 + agency3
       Intrin =~ intrin1 + intrin2 + intrin3
       Extrin =~ extrin1 + extrin2 + extrin3
       PosAff =~ posaff1 + posaff2 + posaff3

       PosAff ~ Agency + Intrin + Extrin'
out2 <- sem(model = m2, data = panas7, std.lv = TRUE, meanstructure = TRUE)
summary(out2, standardized = TRUE, fit.measures = TRUE, rsquare = TRUE)
```

```
## lavaan 0.6-21 ended normally after 57 iterations
##
##   Estimator                                         ML
##   Optimization method                           NLMINB
##   Number of model parameters                        42
##
##   Number of observations                           380
##
## Model Test User Model:
##
##   Test statistic                                57.525
##   Degrees of freedom                                48
##   P-value (Chi-square)                           0.163
##
## Model Test Baseline Model:
##
##   Test statistic                              2806.169
##   Degrees of freedom                                66
##   P-value                                        0.000
##
## User Model versus Baseline Model:
##
##   Comparative Fit Index (CFI)                    0.997
##   Tucker-Lewis Index (TLI)                       0.995
##
## Loglikelihood and Information Criteria:
##
##   Loglikelihood user model (H0)              -2976.044
##   Loglikelihood unrestricted model (H1)      -2947.282
##
##   Akaike (AIC)                                6036.088
##   Bayesian (BIC)                              6201.576
##   Sample-size adjusted Bayesian (SABIC)       6068.318
##
## Root Mean Square Error of Approximation:
##
##   RMSEA                                          0.023
```

```
##   90 Percent confidence interval - lower         0.000
##   90 Percent confidence interval - upper         0.042
##   P-value H_0: RMSEA <= 0.050                    0.992
##   P-value H_0: RMSEA >= 0.080                    0.000
##
## Standardized Root Mean Square Residual:
##
##   SRMR                                           0.026
##
## Parameter Estimates:
##
##   Standard errors                             Standard
##   Information                                 Expected
##   Information saturated (h1) model          Structured
##
## Latent Variables:
##                    Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
##   Agency =~
##     agency1           0.466    0.021   22.559    0.000    0.466    0.905
##     agency2           0.492    0.021   23.764    0.000    0.492    0.933
##     agency3           0.497    0.022   22.826    0.000    0.497    0.911
##   Intrin =~
##     intrin1           0.539    0.039   13.918    0.000    0.539    0.702
##     intrin2           0.579    0.043   13.435    0.000    0.579    0.681
##     intrin3           0.618    0.038   16.272    0.000    0.618    0.803
##   Extrin =~
##     extrin1           0.386    0.022   17.454    0.000    0.386    0.804
##     extrin2           0.456    0.027   17.116    0.000    0.456    0.791
##     extrin3           0.473    0.026   18.204    0.000    0.473    0.831
##   PosAff =~
##     posaff1           0.514    0.026   19.426    0.000    0.570    0.854
##     posaff2           0.545    0.027   20.321    0.000    0.604    0.883
##     posaff3           0.570    0.028   20.682    0.000    0.631    0.894
##
## Regressions:
##                    Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
##   PosAff ~
##     Agency            0.163    0.075    2.158    0.031    0.147    0.147
##     Intrin            0.373    0.082    4.554    0.000    0.337    0.337
##     Extrin           -0.066    0.065   -1.014    0.310   -0.059   -0.059
##
## Covariances:
##                    Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
##   Agency ~~
##     Intrin            0.507    0.047   10.795    0.000    0.507    0.507
##     Extrin            0.270    0.054    5.035    0.000    0.270    0.270
##   Intrin ~~
##     Extrin           -0.027    0.063   -0.433    0.665   -0.027   -0.027
##
## Intercepts:
##                    Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
##    .agency1           2.442    0.026   92.349    0.000    2.442    4.737
##    .agency2           2.550    0.027   94.360    0.000    2.550    4.841
##    .agency3           2.544    0.028   90.926    0.000    2.544    4.664
```

```
##    .intrin1           3.002    0.039   76.152    0.000    3.002    3.907
##    .intrin2           2.987    0.044   68.467    0.000    2.987    3.512
##    .intrin3           3.080    0.039   78.022    0.000    3.080    4.002
##    .extrin1           1.615    0.025   65.603    0.000    1.615    3.365
##    .extrin2           1.686    0.030   57.065    0.000    1.686    2.927
##    .extrin3           1.633    0.029   55.873    0.000    1.633    2.866
##    .posaff1           3.136    0.034   91.663    0.000    3.136    4.702
##    .posaff2           2.991    0.035   85.211    0.000    2.991    4.371
##    .posaff3           3.069    0.036   84.776    0.000    3.069    4.349
##
## Variances:
##                    Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
##    .agency1           0.048    0.005    9.650    0.000    0.048    0.181
##    .agency2           0.036    0.005    7.622    0.000    0.036    0.129
##    .agency3           0.050    0.005    9.260    0.000    0.050    0.169
##    .intrin1           0.300    0.029   10.237    0.000    0.300    0.507
##    .intrin2           0.388    0.036   10.662    0.000    0.388    0.536
##    .intrin3           0.210    0.029    7.346    0.000    0.210    0.356
##    .extrin1           0.081    0.009    9.023    0.000    0.081    0.354
##    .extrin2           0.124    0.013    9.447    0.000    0.124    0.374
##    .extrin3           0.101    0.013    7.997    0.000    0.101    0.310
##    .posaff1           0.120    0.012    9.890    0.000    0.120    0.270
##    .posaff2           0.103    0.012    8.575    0.000    0.103    0.221
##    .posaff3           0.100    0.013    7.939    0.000    0.100    0.201
##     Agency            1.000                               1.000    1.000
##     Intrin            1.000                               1.000    1.000
##     Extrin            1.000                               1.000    1.000
##    .PosAff            1.000                               0.815    0.815
##
## R-Square:
##                    Estimate
##     agency1           0.819
##     agency2           0.871
##     agency3           0.831
##     intrin1           0.493
##     intrin2           0.464
##     intrin3           0.644
##     extrin1           0.646
##     extrin2           0.626
##     extrin3           0.690
##     posaff1           0.730
##     posaff2           0.779
##     posaff3           0.799
##     PosAff            0.185
```

```r
lavaanPlot(model = out2,
           node_options = list(shape = "box", fontname = "Helvetica"),
           coefs = TRUE, covs = TRUE,  stand = TRUE, stars = "regress")
```

```r
### Add negative affect
m3 <- 'Agency =~ agency1 + agency2 + agency3
       Intrin =~ intrin1 + intrin2 + intrin3
       Extrin =~ extrin1 + extrin2 + extrin3
       PosAff =~ posaff1 + posaff2 + posaff3
```

```r
       NegAff =~ negaff1 + negaff2 + negaff3

       PosAff ~ Agency + Intrin + Extrin
       NegAff ~ Agency + Intrin + Extrin'
out3 <- sem(model = m3, data = panas7, std.lv = TRUE, meanstructure = TRUE)
summary(out3, standardized = TRUE, fit.measures = TRUE, rsquare = TRUE)
```

```
## lavaan 0.6-21 ended normally after 61 iterations
##
##   Estimator                                         ML
##   Optimization method                           NLMINB
##   Number of model parameters                        55
##
##   Number of observations                           380
##
## Model Test User Model:
##
##   Test statistic                               106.847
##   Degrees of freedom                                80
##   P-value (Chi-square)                           0.024
##
## Model Test Baseline Model:
##
##   Test statistic                              3749.411
##   Degrees of freedom                               105
##   P-value                                        0.000
##
## User Model versus Baseline Model:
##
##   Comparative Fit Index (CFI)                    0.993
##   Tucker-Lewis Index (TLI)                       0.990
##
## Loglikelihood and Information Criteria:
##
##   Loglikelihood user model (H0)              -3699.110
##   Loglikelihood unrestricted model (H1)      -3645.686
##
##   Akaike (AIC)                                7508.219
##   Bayesian (BIC)                              7724.928
##   Sample-size adjusted Bayesian (SABIC)       7550.424
##
## Root Mean Square Error of Approximation:
##
##   RMSEA                                          0.030
##   90 Percent confidence interval - lower         0.011
##   90 Percent confidence interval - upper         0.044
##   P-value H_0: RMSEA <= 0.050                    0.994
##   P-value H_0: RMSEA >= 0.080                    0.000
##
## Standardized Root Mean Square Residual:
##
##   SRMR                                           0.029
##
```

```
## Parameter Estimates:
##
##   Standard errors                             Standard
##   Information                                 Expected
##   Information saturated (h1) model          Structured
##
## Latent Variables:
##                    Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
##   Agency =~
##     agency1           0.466    0.021   22.560    0.000    0.466    0.905
##     agency2           0.492    0.021   23.774    0.000    0.492    0.934
##     agency3           0.497    0.022   22.815    0.000    0.497    0.911
##   Intrin =~
##     intrin1           0.541    0.039   13.960    0.000    0.541    0.704
##     intrin2           0.581    0.043   13.489    0.000    0.581    0.683
##     intrin3           0.615    0.038   16.201    0.000    0.615    0.799
##   Extrin =~
##     extrin1           0.388    0.022   17.606    0.000    0.388    0.808
##     extrin2           0.456    0.027   17.168    0.000    0.456    0.792
##     extrin3           0.471    0.026   18.100    0.000    0.471    0.826
##   PosAff =~
##     posaff1           0.514    0.026   19.400    0.000    0.569    0.853
##     posaff2           0.544    0.027   20.295    0.000    0.603    0.882
##     posaff3           0.570    0.028   20.717    0.000    0.632    0.895
##   NegAff =~
##     negaff1           0.618    0.029   21.445    0.000    0.634    0.889
##     negaff2           0.571    0.027   21.238    0.000    0.585    0.883
##     negaff3           0.584    0.026   22.536    0.000    0.598    0.918
##
## Regressions:
##                    Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
##   PosAff ~
##     Agency            0.163    0.075    2.158    0.031    0.147    0.147
##     Intrin            0.374    0.082    4.558    0.000    0.338    0.338
##     Extrin           -0.066    0.065   -1.018    0.309   -0.060   -0.060
##   NegAff ~
##     Agency           -0.082    0.073   -1.113    0.266   -0.080   -0.080
##     Intrin            0.069    0.076    0.908    0.364    0.067    0.067
##     Extrin            0.238    0.065    3.692    0.000    0.232    0.232
##
## Covariances:
##                    Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
##   Agency ~~
##     Intrin            0.507    0.047   10.782    0.000    0.507    0.507
##     Extrin            0.270    0.054    5.028    0.000    0.270    0.270
##   Intrin ~~
##     Extrin           -0.028    0.063   -0.441    0.659   -0.028   -0.028
##  .PosAff ~~
##    .NegAff           -0.077    0.058   -1.338    0.181   -0.077   -0.077
##
## Intercepts:
##                    Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
##    .agency1           2.442    0.026   92.349    0.000    2.442    4.737
##    .agency2           2.550    0.027   94.360    0.000    2.550    4.841
```

```
##    .agency3           2.544    0.028   90.926    0.000    2.544    4.664
##    .intrin1           3.002    0.039   76.152    0.000    3.002    3.907
##    .intrin2           2.987    0.044   68.467    0.000    2.987    3.512
##    .intrin3           3.080    0.039   78.022    0.000    3.080    4.002
##    .extrin1           1.615    0.025   65.603    0.000    1.615    3.365
##    .extrin2           1.686    0.030   57.065    0.000    1.686    2.927
##    .extrin3           1.633    0.029   55.873    0.000    1.633    2.866
##    .posaff1           3.136    0.034   91.663    0.000    3.136    4.702
##    .posaff2           2.991    0.035   85.211    0.000    2.991    4.371
##    .posaff3           3.069    0.036   84.776    0.000    3.069    4.349
##    .negaff1           1.701    0.037   46.481    0.000    1.701    2.384
##    .negaff2           1.527    0.034   44.944    0.000    1.527    2.306
##    .negaff3           1.545    0.033   46.195    0.000    1.545    2.370
##
## Variances:
##                    Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
##    .agency1           0.048    0.005    9.649    0.000    0.048    0.181
##    .agency2           0.036    0.005    7.604    0.000    0.036    0.129
##    .agency3           0.051    0.005    9.278    0.000    0.051    0.170
##    .intrin1           0.298    0.029   10.204    0.000    0.298    0.505
##    .intrin2           0.386    0.036   10.620    0.000    0.386    0.533
##    .intrin3           0.214    0.029    7.474    0.000    0.214    0.361
##    .extrin1           0.080    0.009    8.924    0.000    0.080    0.347
##    .extrin2           0.123    0.013    9.465    0.000    0.123    0.372
##    .extrin3           0.103    0.012    8.264    0.000    0.103    0.318
##    .posaff1           0.121    0.012    9.921    0.000    0.121    0.272
##    .posaff2           0.104    0.012    8.615    0.000    0.104    0.222
##    .posaff3           0.099    0.013    7.870    0.000    0.099    0.198
##    .negaff1           0.107    0.012    9.202    0.000    0.107    0.210
##    .negaff2           0.096    0.010    9.498    0.000    0.096    0.220
##    .negaff3           0.067    0.009    7.369    0.000    0.067    0.158
##     Agency            1.000                                1.000    1.000
##     Intrin            1.000                                1.000    1.000
##     Extrin            1.000                                1.000    1.000
##    .PosAff            1.000                                0.814    0.814
##    .NegAff            1.000                                0.951    0.951
##
## R-Square:
##                    Estimate
##     agency1           0.819
##     agency2           0.871
##     agency3           0.830
##     intrin1           0.495
##     intrin2           0.467
##     intrin3           0.639
##     extrin1           0.653
##     extrin2           0.628
##     extrin3           0.682
##     posaff1           0.728
##     posaff2           0.778
##     posaff3           0.802
##     negaff1           0.790
##     negaff2           0.780
##     negaff3           0.842
```

```
##     PosAff            0.186
##     NegAff            0.049
```

```r
lavaanPlot(model = out3,
           node_options = list(shape = "box", fontname = "Helvetica"),
           coefs = TRUE, covs = TRUE,  stand = TRUE, stars = "regress")
```

2.3. Prune the Model

```r
m4 <- 'Agency =~ agency1 + agency2 + agency3
       Intrin =~ intrin1 + intrin2 + intrin3
       Extrin =~ extrin1 + extrin2 + extrin3
       PosAff =~ posaff1 + posaff2 + posaff3
       NegAff =~ negaff1 + negaff2 + negaff3

       PosAff ~ Agency + Intrin
       NegAff ~ Extrin'
out4 <- sem(model = m4, data = panas7, std.lv = TRUE, meanstructure = TRUE)
summary(out4, standardized = TRUE, fit.measures = TRUE, rsquare = TRUE)
```

```
## lavaan 0.6-21 ended normally after 58 iterations
##
##   Estimator                                         ML
##   Optimization method                           NLMINB
##   Number of model parameters                        52
##
##   Number of observations                           380
##
## Model Test User Model:
##
##   Test statistic                               109.152
##   Degrees of freedom                                83
##   P-value (Chi-square)                           0.029
##
## Model Test Baseline Model:
##
##   Test statistic                              3749.411
##   Degrees of freedom                               105
##   P-value                                        0.000
##
## User Model versus Baseline Model:
##
##   Comparative Fit Index (CFI)                    0.993
##   Tucker-Lewis Index (TLI)                       0.991
##
## Loglikelihood and Information Criteria:
##
##   Loglikelihood user model (H0)              -3700.262
##   Loglikelihood unrestricted model (H1)      -3645.686
##
##   Akaike (AIC)                                7504.524
```

```
##   Bayesian (BIC)                              7709.413
##   Sample-size adjusted Bayesian (SABIC)       7544.427
##
## Root Mean Square Error of Approximation:
##
##   RMSEA                                          0.029
##   90 Percent confidence interval - lower         0.010
##   90 Percent confidence interval - upper         0.043
##   P-value H_0: RMSEA <= 0.050                    0.996
##   P-value H_0: RMSEA >= 0.080                    0.000
##
## Standardized Root Mean Square Residual:
##
##   SRMR                                           0.032
##
## Parameter Estimates:
##
##   Standard errors                             Standard
##   Information                                 Expected
##   Information saturated (h1) model          Structured
##
## Latent Variables:
##                    Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
##   Agency =~
##     agency1           0.466    0.021   22.556    0.000    0.466    0.905
##     agency2           0.492    0.021   23.759    0.000    0.492    0.933
##     agency3           0.497    0.022   22.835    0.000    0.497    0.912
##   Intrin =~
##     intrin1           0.540    0.039   13.950    0.000    0.540    0.703
##     intrin2           0.580    0.043   13.454    0.000    0.580    0.682
##     intrin3           0.616    0.038   16.235    0.000    0.616    0.800
##   Extrin =~
##     extrin1           0.388    0.022   17.581    0.000    0.388    0.808
##     extrin2           0.457    0.027   17.168    0.000    0.457    0.793
##     extrin3           0.471    0.026   18.116    0.000    0.471    0.827
##   PosAff =~
##     posaff1           0.514    0.027   19.371    0.000    0.569    0.854
##     posaff2           0.545    0.027   20.264    0.000    0.604    0.882
##     posaff3           0.570    0.028   20.658    0.000    0.632    0.895
##   NegAff =~
##     negaff1           0.620    0.029   21.499    0.000    0.633    0.889
##     negaff2           0.573    0.027   21.311    0.000    0.585    0.883
##     negaff3           0.586    0.026   22.616    0.000    0.598    0.918
##
## Regressions:
##                    Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
##   PosAff ~
##     Agency            0.129    0.070    1.847    0.065    0.116    0.116
##     Intrin            0.398    0.080    4.962    0.000    0.359    0.359
##   NegAff ~
##     Extrin            0.207    0.059    3.496    0.000    0.202    0.202
##
## Covariances:
##                    Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
```

```
##   Agency ~~
##     Intrin            0.507    0.047   10.796    0.000    0.507    0.507
##     Extrin            0.268    0.054    4.996    0.000    0.268    0.268
##   Intrin ~~
##     Extrin           -0.032    0.062   -0.510    0.610   -0.032   -0.032
##  .PosAff ~~
##    .NegAff           -0.076    0.057   -1.334    0.182   -0.076   -0.076
##
## Intercepts:
##                    Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
##    .agency1           2.442    0.026   92.349    0.000    2.442    4.737
##    .agency2           2.550    0.027   94.360    0.000    2.550    4.841
##    .agency3           2.544    0.028   90.926    0.000    2.544    4.664
##    .intrin1           3.002    0.039   76.152    0.000    3.002    3.907
##    .intrin2           2.987    0.044   68.467    0.000    2.987    3.512
##    .intrin3           3.080    0.039   78.022    0.000    3.080    4.002
##    .extrin1           1.615    0.025   65.603    0.000    1.615    3.365
##    .extrin2           1.686    0.030   57.065    0.000    1.686    2.927
##    .extrin3           1.633    0.029   55.873    0.000    1.633    2.866
##    .posaff1           3.136    0.034   91.659    0.000    3.136    4.702
##    .posaff2           2.991    0.035   85.207    0.000    2.991    4.371
##    .posaff3           3.069    0.036   84.772    0.000    3.069    4.349
##    .negaff1           1.701    0.037   46.504    0.000    1.701    2.386
##    .negaff2           1.527    0.034   44.966    0.000    1.527    2.307
##    .negaff3           1.545    0.033   46.219    0.000    1.545    2.371
##
## Variances:
##                    Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
##    .agency1           0.048    0.005    9.652    0.000    0.048    0.181
##    .agency2           0.036    0.005    7.627    0.000    0.036    0.129
##    .agency3           0.050    0.005    9.244    0.000    0.050    0.169
##    .intrin1           0.299    0.029   10.225    0.000    0.299    0.506
##    .intrin2           0.387    0.036   10.659    0.000    0.387    0.535
##    .intrin3           0.213    0.029    7.452    0.000    0.213    0.359
##    .extrin1           0.080    0.009    8.937    0.000    0.080    0.348
##    .extrin2           0.123    0.013    9.449    0.000    0.123    0.372
##    .extrin3           0.103    0.013    8.217    0.000    0.103    0.317
##    .posaff1           0.121    0.012    9.911    0.000    0.121    0.271
##    .posaff2           0.104    0.012    8.597    0.000    0.104    0.222
##    .posaff3           0.099    0.013    7.895    0.000    0.099    0.199
##    .negaff1           0.107    0.012    9.216    0.000    0.107    0.210
##    .negaff2           0.096    0.010    9.482    0.000    0.096    0.219
##    .negaff3           0.067    0.009    7.362    0.000    0.067    0.158
##     Agency            1.000                                1.000    1.000
##     Intrin            1.000                                1.000    1.000
##     Extrin            1.000                                1.000    1.000
##    .PosAff            1.000                                0.815    0.815
##    .NegAff            1.000                                0.959    0.959
##
## R-Square:
##                    Estimate
##     agency1           0.819
##     agency2           0.871
##     agency3           0.831
```

```
##     intrin1           0.494
##     intrin2           0.465
##     intrin3           0.641
##     extrin1           0.652
##     extrin2           0.628
##     extrin3           0.683
##     posaff1           0.729
##     posaff2           0.778
##     posaff3           0.801
##     negaff1           0.790
##     negaff2           0.781
##     negaff3           0.842
##     PosAff            0.185
##     NegAff            0.041
```

```
lavaanPlot(model = out4,
           node_options = list(shape = "box", fontname = "Helvetica"),
           coefs = TRUE, covs = TRUE,  stand = TRUE, stars = "regress")
```

Likelihood-ratio test

Likelihood-ratio test (LRT) is a statistical method used for comparing two nested statistical models, typically in the context of hypothesis testing. It is a common tool in statistics and econometrics and is used to determine whether a more complex model significantly improves the fit of the data compared to a simpler model. LRT is based on the likelihood function, which measures how well a statistical model explains the observed data.

- The likelihood ratio statistic (LR) is calculated as the ratio of the likelihood of the data under the more complex model (H1) to the likelihood of the data under the simpler model (H0).
- If LR is greater than the critical value, you reject the null hypothesis (H0) in favor of the alternative hypothesis (H1), indicating that the more complex model provides a significantly better fit to the data.
- If LR is not greater than the critical value, you fail to reject the null hypothesis, suggesting that the simpler model is sufficient to explain the data.

```
anova(out3, out4)
```

```
##
## Chi-Squared Difference Test
##
##      Df    AIC    BIC  Chisq Chisq diff RMSEA Df diff Pr(>Chisq)
## out3 80 7508.2 7724.9 106.85
## out4 83 7504.5 7709.4 109.15     2.3051     0       3     0.5116
```

3. Mediation

Mediation refers to a statistical relationship between three variables, where one variable (the mediator) helps explain or account for the relationship between two other variables (the independent variable and the dependent variable).

Mediation analysis aims to decompose the total effect of the independent variable (IV) on the dependent variable (DV) into two components:

- Direct effect: This is the part of the relationship between the independent variable (IV) and the dependent variable (DV) that remains after accounting for the mediator (M). In other words, it is the effect of the independent variable on the dependent variable that is not mediated by the mediator.
- Indirect effect: This is the part of the relationship between the independent variable (IV) and the dependent variable (DV) that is mediated or explained by the mediator (M). It quantifies the extent to which the mediator helps explain the relationship between the IV and DV.

3.1. Steps

```r
### Step 0: Show that X affects Y
m5 <- 'Intrin =~ intrin1 + intrin2 + intrin3
       PosAff =~ posaff1 + posaff2 + posaff3

       PosAff ~ Intrin'
out5 <- sem(model = m5, data = panas7, std.lv = TRUE, meanstructure = TRUE)
summary(out5, standardized = TRUE, fit.measures = TRUE, rsquare = TRUE)
```

```
## lavaan 0.6-21 ended normally after 27 iterations
##
##   Estimator                                         ML
##   Optimization method                           NLMINB
##   Number of model parameters                        19
##
##   Number of observations                           380
##
## Model Test User Model:
##
##   Test statistic                                 8.508
##   Degrees of freedom                                 8
##   P-value (Chi-square)                           0.385
##
## Model Test Baseline Model:
##
##   Test statistic                              1124.549
##   Degrees of freedom                                15
##   P-value                                        0.000
##
## User Model versus Baseline Model:
##
##   Comparative Fit Index (CFI)                    1.000
##   Tucker-Lewis Index (TLI)                       0.999
##
## Loglikelihood and Information Criteria:
##
##   Loglikelihood user model (H0)              -1985.326
##   Loglikelihood unrestricted model (H1)      -1981.072
##
##   Akaike (AIC)                                4008.652
##   Bayesian (BIC)                              4083.515
##   Sample-size adjusted Bayesian (SABIC)       4023.232
##
```

```
## Root Mean Square Error of Approximation:
##
##   RMSEA                                          0.013
##   90 Percent confidence interval - lower         0.000
##   90 Percent confidence interval - upper         0.062
##   P-value H_0: RMSEA <= 0.050                    0.862
##   P-value H_0: RMSEA >= 0.080                    0.007
##
## Standardized Root Mean Square Residual:
##
##   SRMR                                           0.021
##
## Parameter Estimates:
##
##   Standard errors                             Standard
##   Information                                 Expected
##   Information saturated (h1) model          Structured
##
## Latent Variables:
##                    Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
##   Intrin =~
##     intrin1           0.547    0.040   13.843    0.000    0.547    0.712
##     intrin2           0.582    0.044   13.267    0.000    0.582    0.685
##     intrin3           0.608    0.039   15.447    0.000    0.608    0.790
##   PosAff =~
##     posaff1           0.518    0.027   19.242    0.000    0.570    0.854
##     posaff2           0.550    0.027   20.146    0.000    0.605    0.884
##     posaff3           0.574    0.028   20.428    0.000    0.630    0.893
##
## Regressions:
##                    Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
##   PosAff ~
##     Intrin            0.456    0.069    6.566    0.000    0.415    0.415
##
## Intercepts:
##                    Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
##    .intrin1           3.002    0.039   76.152    0.000    3.002    3.907
##    .intrin2           2.987    0.044   68.467    0.000    2.987    3.512
##    .intrin3           3.080    0.039   78.022    0.000    3.080    4.002
##    .posaff1           3.136    0.034   91.663    0.000    3.136    4.702
##    .posaff2           2.991    0.035   85.211    0.000    2.991    4.371
##    .posaff3           3.069    0.036   84.776    0.000    3.069    4.349
##
## Variances:
##                    Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
##    .intrin1           0.291    0.030    9.539    0.000    0.291    0.493
##    .intrin2           0.384    0.038   10.222    0.000    0.384    0.531
##    .intrin3           0.222    0.031    7.158    0.000    0.222    0.376
##    .posaff1           0.120    0.012    9.874    0.000    0.120    0.270
##    .posaff2           0.102    0.012    8.488    0.000    0.102    0.219
##    .posaff3           0.101    0.013    7.974    0.000    0.101    0.203
##     Intrin            1.000                               1.000    1.000
##    .PosAff            1.000                               0.828    0.828
##
```

```
## R-Square:
##                    Estimate
##     intrin1           0.507
##     intrin2           0.469
##     intrin3           0.624
##     posaff1           0.730
##     posaff2           0.781
##     posaff3           0.797
##     PosAff            0.172
```

```r
### Step 1: Show that X affects M
m6 <- 'Agency =~ agency1 + agency2 + agency3
       Intrin =~ intrin1 + intrin2 + intrin3

       Agency ~ Intrin'
out6 <- sem(model = m6, data = panas7, std.lv = TRUE, meanstructure = TRUE)
summary(out6, standardized = TRUE, fit.measures = TRUE, rsquare = TRUE)
```

```
## lavaan 0.6-21 ended normally after 30 iterations
##
##   Estimator                                         ML
##   Optimization method                           NLMINB
##   Number of model parameters                        19
##
##   Number of observations                           380
##
## Model Test User Model:
##
##   Test statistic                                10.708
##   Degrees of freedom                                 8
##   P-value (Chi-square)                           0.219
##
## Model Test Baseline Model:
##
##   Test statistic                              1416.263
##   Degrees of freedom                                15
##   P-value                                        0.000
##
## User Model versus Baseline Model:
##
##   Comparative Fit Index (CFI)                    0.998
##   Tucker-Lewis Index (TLI)                       0.996
##
## Loglikelihood and Information Criteria:
##
##   Loglikelihood user model (H0)              -1545.377
##   Loglikelihood unrestricted model (H1)      -1540.023
##
##   Akaike (AIC)                                3128.754
##   Bayesian (BIC)                              3203.617
##   Sample-size adjusted Bayesian (SABIC)       3143.334
##
## Root Mean Square Error of Approximation:
##
```

```
##   RMSEA                                          0.030
##   90 Percent confidence interval - lower         0.000
##   90 Percent confidence interval - upper         0.071
##   P-value H_0: RMSEA <= 0.050                    0.745
##   P-value H_0: RMSEA >= 0.080                    0.019
##
## Standardized Root Mean Square Residual:
##
##   SRMR                                           0.016
##
## Parameter Estimates:
##
##   Standard errors                             Standard
##   Information                                 Expected
##   Information saturated (h1) model          Structured
##
## Latent Variables:
##                    Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
##   Agency =~
##     agency1           0.401    0.019   20.727    0.000    0.465    0.903
##     agency2           0.424    0.020   21.702    0.000    0.492    0.934
##     agency3           0.429    0.020   21.034    0.000    0.498    0.913
##   Intrin =~
##     intrin1           0.529    0.039   13.477    0.000    0.529    0.688
##     intrin2           0.579    0.044   13.314    0.000    0.579    0.681
##     intrin3           0.626    0.039   16.232    0.000    0.626    0.814
##
## Regressions:
##                    Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
##   Agency ~
##     Intrin            0.588    0.073    8.045    0.000    0.507    0.507
##
## Intercepts:
##                    Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
##    .agency1           2.442    0.026   92.349    0.000    2.442    4.737
##    .agency2           2.550    0.027   94.360    0.000    2.550    4.841
##    .agency3           2.544    0.028   90.926    0.000    2.544    4.664
##    .intrin1           3.002    0.039   76.152    0.000    3.002    3.907
##    .intrin2           2.987    0.044   68.467    0.000    2.987    3.512
##    .intrin3           3.080    0.039   78.022    0.000    3.080    4.002
##
## Variances:
##                    Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
##    .agency1           0.049    0.005    9.703    0.000    0.049    0.185
##    .agency2           0.036    0.005    7.506    0.000    0.036    0.128
##    .agency3           0.050    0.005    9.114    0.000    0.050    0.167
##    .intrin1           0.311    0.030   10.321    0.000    0.311    0.526
##    .intrin2           0.388    0.037   10.480    0.000    0.388    0.536
##    .intrin3           0.200    0.030    6.606    0.000    0.200    0.337
##    .Agency            1.000                               0.743    0.743
##     Intrin            1.000                               1.000    1.000
##
## R-Square:
##                    Estimate
```

```
##     agency1           0.815
##     agency2           0.872
##     agency3           0.833
##     intrin1           0.474
##     intrin2           0.464
##     intrin3           0.663
##     Agency            0.257
```

```r
### Step 2: Show that M affects Y when controlling for X
### Step 3: Determine whether mediation is full or partial
m7 <- 'Agency =~ agency1 + agency2 + agency3
       Intrin =~ intrin1 + intrin2 + intrin3
       PosAff =~ posaff1 + posaff2 + posaff3

       Agency ~ Intrin
       PosAff ~ Agency + Intrin'
out7 <- sem(model = m7, data = panas7, std.lv = TRUE, meanstructure = TRUE)
summary(out7, standardized = TRUE, fit.measures = TRUE, rsquare = TRUE)
```

```
## lavaan 0.6-21 ended normally after 44 iterations
##
##   Estimator                                         ML
##   Optimization method                           NLMINB
##   Number of model parameters                        30
##
##   Number of observations                           380
##
## Model Test User Model:
##
##   Test statistic                                22.377
##   Degrees of freedom                                24
##   P-value (Chi-square)                           0.557
##
## Model Test Baseline Model:
##
##   Test statistic                              2248.686
##   Degrees of freedom                                36
##   P-value                                        0.000
##
## User Model versus Baseline Model:
##
##   Comparative Fit Index (CFI)                    1.000
##   Tucker-Lewis Index (TLI)                       1.001
##
## Loglikelihood and Information Criteria:
##
##   Loglikelihood user model (H0)              -2321.962
##   Loglikelihood unrestricted model (H1)      -2310.773
##
##   Akaike (AIC)                                4703.923
##   Bayesian (BIC)                              4822.128
##   Sample-size adjusted Bayesian (SABIC)       4726.944
##
## Root Mean Square Error of Approximation:
```

```
##
##   RMSEA                                          0.000
##   90 Percent confidence interval - lower         0.000
##   90 Percent confidence interval - upper         0.038
##   P-value H_0: RMSEA <= 0.050                    0.993
##   P-value H_0: RMSEA >= 0.080                    0.000
##
## Standardized Root Mean Square Residual:
##
##   SRMR                                           0.021
##
## Parameter Estimates:
##
##   Standard errors                             Standard
##   Information                                 Expected
##   Information saturated (h1) model          Structured
##
## Latent Variables:
##                    Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
##   Agency =~
##     agency1           0.401    0.019   20.701    0.000    0.465    0.902
##     agency2           0.424    0.020   21.713    0.000    0.492    0.934
##     agency3           0.429    0.020   21.024    0.000    0.498    0.913
##   Intrin =~
##     intrin1           0.537    0.039   13.833    0.000    0.537    0.699
##     intrin2           0.576    0.043   13.322    0.000    0.576    0.677
##     intrin3           0.622    0.038   16.356    0.000    0.622    0.808
##   PosAff =~
##     posaff1           0.515    0.027   19.429    0.000    0.570    0.854
##     posaff2           0.546    0.027   20.326    0.000    0.604    0.883
##     posaff3           0.570    0.028   20.654    0.000    0.631    0.893
##
## Regressions:
##                    Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
##   Agency ~
##     Intrin            0.588    0.073    8.038    0.000    0.507    0.507
##   PosAff ~
##     Agency            0.119    0.061    1.965    0.049    0.125    0.125
##     Intrin            0.386    0.080    4.821    0.000    0.349    0.349
##
## Intercepts:
##                    Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
##    .agency1           2.442    0.026   92.349    0.000    2.442    4.737
##    .agency2           2.550    0.027   94.360    0.000    2.550    4.841
##    .agency3           2.544    0.028   90.926    0.000    2.544    4.664
##    .intrin1           3.002    0.039   76.152    0.000    3.002    3.907
##    .intrin2           2.987    0.044   68.467    0.000    2.987    3.512
##    .intrin3           3.080    0.039   78.022    0.000    3.080    4.002
##    .posaff1           3.136    0.034   91.663    0.000    3.136    4.702
##    .posaff2           2.991    0.035   85.211    0.000    2.991    4.371
##    .posaff3           3.069    0.036   84.776    0.000    3.069    4.349
##
## Variances:
##                    Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
```

```
##    .agency1           0.049    0.005    9.748    0.000    0.049    0.186
##    .agency2           0.035    0.005    7.472    0.000    0.035    0.127
##    .agency3           0.050    0.005    9.132    0.000    0.050    0.167
##    .intrin1           0.302    0.029   10.262    0.000    0.302    0.511
##    .intrin2           0.392    0.037   10.714    0.000    0.392    0.542
##    .intrin3           0.205    0.029    7.087    0.000    0.205    0.346
##    .posaff1           0.120    0.012    9.876    0.000    0.120    0.270
##    .posaff2           0.103    0.012    8.549    0.000    0.103    0.220
##    .posaff3           0.101    0.013    7.972    0.000    0.101    0.202
##    .Agency            1.000                               0.743    0.743
##     Intrin            1.000                               1.000    1.000
##    .PosAff            1.000                               0.818    0.818
##
## R-Square:
##                    Estimate
##     agency1           0.814
##     agency2           0.873
##     agency3           0.833
##     intrin1           0.489
##     intrin2           0.458
##     intrin3           0.654
##     posaff1           0.730
##     posaff2           0.780
##     posaff3           0.798
##     Agency            0.257
##     PosAff            0.182
```

```r
lavaanPlot(model = out7,
           node_options = list(shape = "box", fontname = "Helvetica"),
           coefs = TRUE, covs = TRUE,  stand = TRUE, stars = "regress")
```

3.2. Bootstrapping

Bootstrapping is a resampling technique in statistics that is used to estimate the sampling distribution of a statistic by repeatedly drawing random samples (with replacement) from the observed data.

Here are the key steps and concepts involved in bootstrapping:

1) Sample data: Start with an observed dataset that contains your sample of interest. This sample can be from any population, and it's the data you want to use to make inferences about population parameters.

2) Resampling: Bootstrapping involves repeatedly drawing random samples from the observed data with replacement. Each resampled dataset is of the same size as the original dataset, but it is created by randomly selecting data points from the original dataset, allowing for the same data point to be selected multiple times while others may not be selected at all.

3) Statistic calculation: Calculate the statistic of interest (e.g., mean, median, standard deviation, regression coefficient, etc.) for each of the resampled datasets. This gives you a collection of statistics that represent the sampling distribution of the statistic.

4) Analyze results: Examine the distribution of the bootstrapped statistics to make inferences about the population parameter of interest. You can construct confidence intervals, estimate standard errors, and perform hypothesis tests based on the bootstrapped results.

```r
m8 <- 'Agency =~ agency1 + agency2 + agency3
       Intrin =~ intrin1 + intrin2 + intrin3
       PosAff =~ posaff1 + posaff2 + posaff3

       Agency ~ a*Intrin
       PosAff ~ b*Agency
       PosAff ~ c*Intrin

       ab := a*b
       total := c+(a*b)'
out8 <- sem(model = m8, data = panas7, std.lv = TRUE, meanstructure = TRUE, se = "bootstrap", bootstrap
parameterEstimates(out8, level = 0.95, boot.ci.type = "bca.simple")
```

```
##        lhs op     rhs label   est    se      z pvalue ci.lower ci.upper
## 1   Agency =~ agency1       0.401 0.022 18.028  0.000    0.358    0.445
## 2   Agency =~ agency2       0.424 0.024 17.612  0.000    0.379    0.472
## 3   Agency =~ agency3       0.429 0.024 17.845  0.000    0.382    0.477
## 4   Intrin =~ intrin1       0.537 0.038 14.078  0.000    0.462    0.613
## 5   Intrin =~ intrin2       0.576 0.037 15.660  0.000    0.502    0.647
## 6   Intrin =~ intrin3       0.622 0.038 16.518  0.000    0.546    0.693
## 7   PosAff =~ posaff1       0.515 0.025 20.384  0.000    0.469    0.568
## 8   PosAff =~ posaff2       0.546 0.029 19.139  0.000    0.494    0.602
## 9   PosAff =~ posaff3       0.570 0.028 20.056  0.000    0.517    0.627
## 10  Agency  ~  Intrin     a 0.588 0.086  6.874  0.000    0.429    0.773
## 11  PosAff  ~  Agency     b 0.119 0.060  1.985  0.047    0.002    0.240
## 12  PosAff  ~  Intrin     c 0.386 0.086  4.473  0.000    0.221    0.559
## 13 agency1 ~~ agency1       0.049 0.006  8.669  0.000    0.039    0.061
## 14 agency2 ~~ agency2       0.035 0.005  7.375  0.000    0.026    0.045
## 15 agency3 ~~ agency3       0.050 0.006  8.620  0.000    0.039    0.062
## 16 intrin1 ~~ intrin1       0.302 0.033  9.167  0.000    0.241    0.371
## 17 intrin2 ~~ intrin2       0.392 0.039 10.054  0.000    0.319    0.474
## 18 intrin3 ~~ intrin3       0.205 0.038  5.413  0.000    0.139    0.288
## 19 posaff1 ~~ posaff1       0.120 0.016  7.675  0.000    0.092    0.153
## 20 posaff2 ~~ posaff2       0.103 0.015  6.771  0.000    0.075    0.134
## 21 posaff3 ~~ posaff3       0.101 0.014  7.273  0.000    0.075    0.129
## 22  Agency ~~  Agency       1.000 0.000     NA     NA    1.000    1.000
## 23  Intrin ~~  Intrin       1.000 0.000     NA     NA    1.000    1.000
## 24  PosAff ~~  PosAff       1.000 0.000     NA     NA    1.000    1.000
## 25 agency1 ~1               2.442 0.027 91.384  0.000    2.389    2.493
## 26 agency2 ~1               2.550 0.027 95.461  0.000    2.498    2.602
## 27 agency3 ~1               2.544 0.028 89.812  0.000    2.487    2.598
## 28 intrin1 ~1               3.002 0.040 75.645  0.000    2.923    3.078
## 29 intrin2 ~1               2.987 0.043 69.911  0.000    2.904    3.071
## 30 intrin3 ~1               3.080 0.039 78.015  0.000    2.999    3.157
## 31 posaff1 ~1               3.136 0.034 91.946  0.000    3.069    3.204
## 32 posaff2 ~1               2.991 0.035 85.222  0.000    2.927    3.065
## 33 posaff3 ~1               3.069 0.036 84.432  0.000    2.999    3.144
## 34  Agency ~1               0.000 0.000     NA     NA    0.000    0.000
## 35  Intrin ~1               0.000 0.000     NA     NA    0.000    0.000
## 36  PosAff ~1               0.000 0.000     NA     NA    0.000    0.000
## 37      ab :=     a*b    ab 0.070 0.039  1.806  0.071    0.002    0.157
## 38   total := c+(a*b) total 0.456 0.070  6.471  0.000    0.321    0.601
```